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Healthcare Market and Industry Perspectives

Moderator: Jane Day
Market Development Manager

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The Palm Healthcare Economy



Palm OS: The Leader in Healthcare

- *"Ease of use, small size, and long battery life [rank] most important" among physicians.*

  —Current Usage Patterns and Attitudes Toward Handheld Computers in Clinical Care, Churchill, et al.

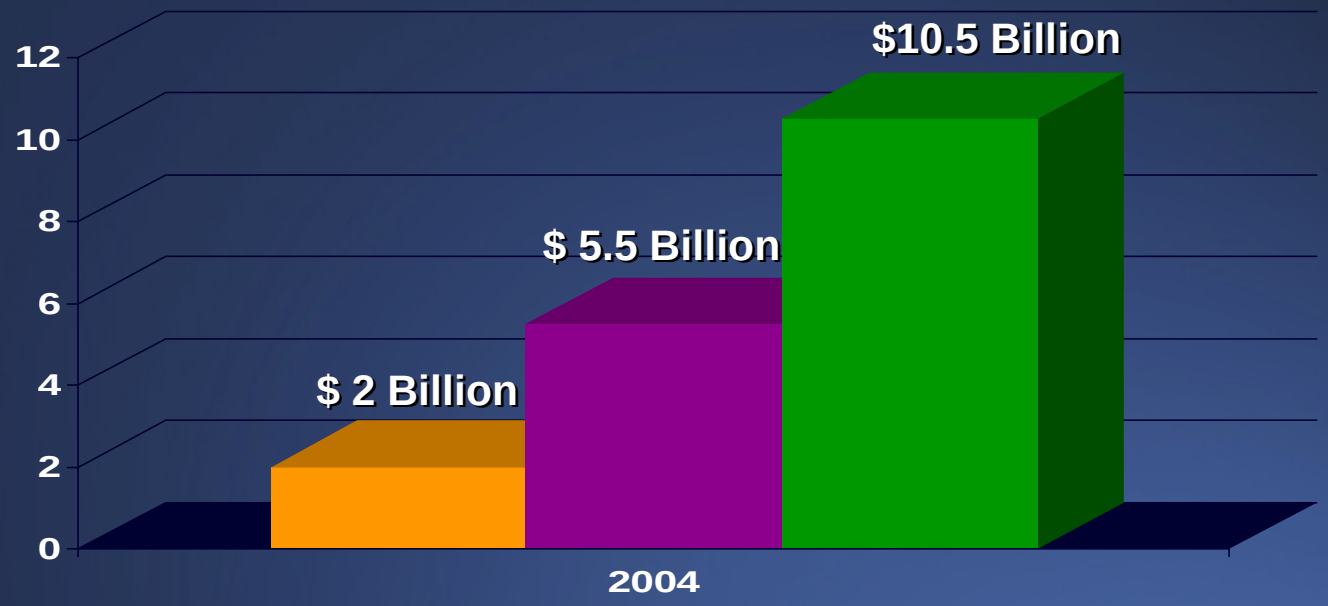
- *"…The Palm OS is by far the leading choice of consumers…This market dominance will undoubtedly continue through the foreseeable future, due to abundance of software and hardware choices available on the Palm OS platform."*

  —Dorenfest Research Note

- *"…Palm's operating system has certain inherent advantages, since it was designed for a handheld device as opposed to Microsoft's who scaled down their Windows operating system to fit on a handheld."*

  —Dorenfest Research Note

Market Opportunity Projections



- *"By 2004 20% of US Physicians will be using handheld devices for transactions, generating over $2 Billion in sales for handheld companies."*
- *"Handheld electronic clinical market (ePrescribing, eLabs) could be a $5.5 Billion market by 2004."*
- *"Total addressable market for handheld companies to be $10.5 billion by 2004."*

—WR Hambrecht & Co.

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Today's Panelists

- **Richard Fiedotin**

  ePocrates, Co-Founder and Vice President

- **Scott Stuewe**

  Cerner Corporation, Lead Partner

- **Cameron Lewis**

  Virtmed, Vice President Business Development

Why Doctors Love Their Palm™ Handhelds

Richard Fiedotin, MD
Vice-President, Business Development
ePocrates

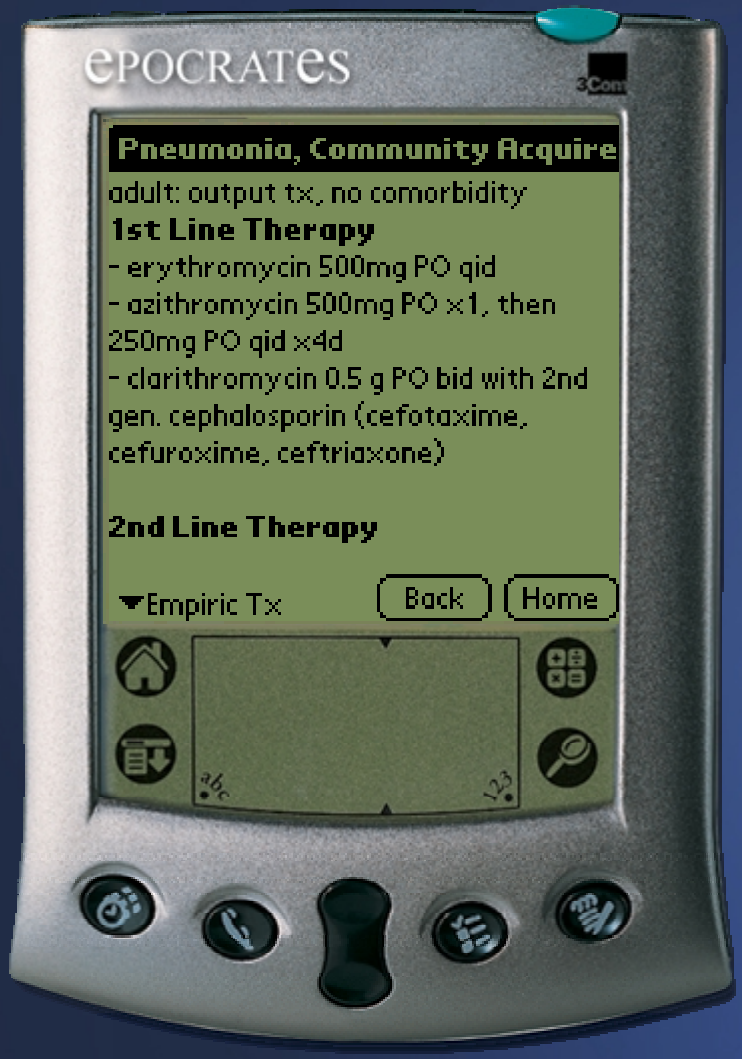
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Three Issues to Address Today

- Applications which doctors value most
- Demographics of early-majority physicians
  - Geography
  - Age
  - Specialty
- Design principles to consider when developing for MDs

Clinical References

qRx and qID: Our Flagship Products



qRx Drug Reference

- 2,300+ drugs
- Updates automatically via Internet
- Harvard Study (pub. in Nov): Averts 1–2 ADE's/MD/week

qID Infectious Disease Guide

- Over 500 disease entities
- Searchable by "bug or drug"
- Directly links to qRx for drug information

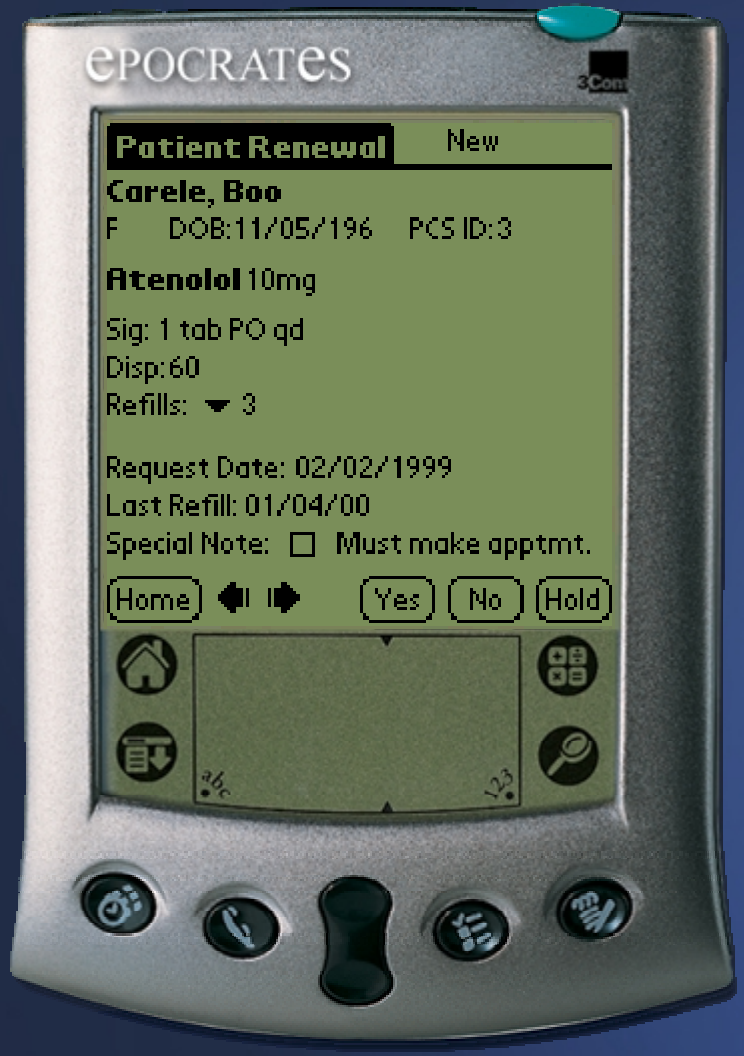
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ePocrates Network Adoption

Palm Platform Driving Our Success

- Over 150,000 users in 13 mo.
  - Adding 500+ new users/day
  - 13+% of US MDs (AMA verified)
  - Avg usage: 12 drug lookups and 1.3 syncs per week
  - 70% word-of-mouth
  - 29% purchase Palms to use qRx
- "A sleeper hit with doctors" —*Wall Street Journal*
- "ePocrates is now the Web's most popular health care destination for doctors." —*AMA*

Prescribing Tools
qRx Pro and qScript



qRx Pro Formulary Reference

- Quick determination of patient medication benefits

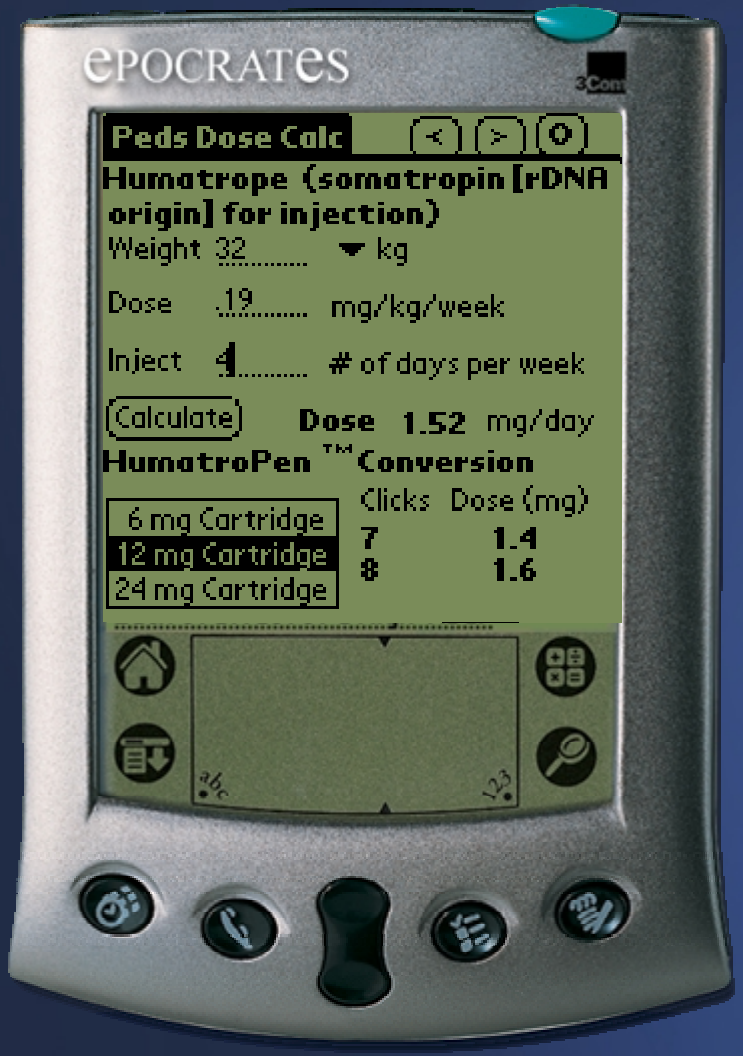
qScript Rx Renewal Tool

- 1-click
- No office integration req'd
- Certicom ECC and SSL

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Content Channels

HTML Browser and Custom Apps



Content Channels

- NEJM, JACC, etc.
- Viewed ~1.4x per week
- Shown to increase subscriptions

Sponsored Applications

- “Beamable” information tools for physicians
- e.g., Eli Lilly Time Action Curves and Product Dosing

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Handheld Messaging
DocAlerts and SyncSpots



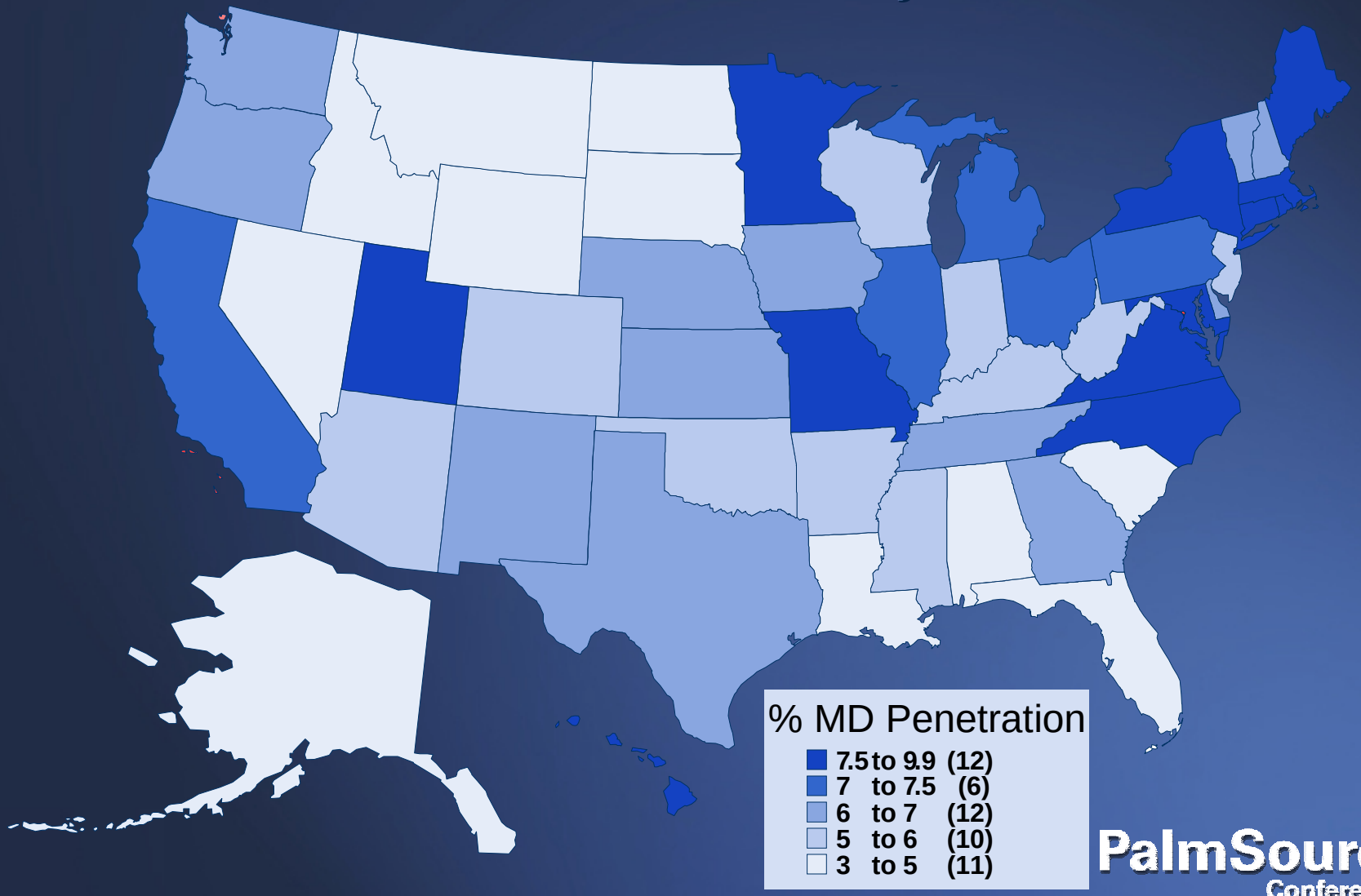
Handheld Messaging
- Appears on handheld following HotSync® operation
- Bi-directional
- Provides clinical updates and invitations

Desktop Messaging
- Appears on desktop during HotSync® operation
- Click-through link to web

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Where Are the Doctors?

At Least 3% of MDs in Every State



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Which Specialties Use Palm™ Handhelds?

ePocrates Network Penetration by Specialty

Physician Penetration by Specialty: Updated September 18, 2000

Specialty		Registered	
	AMA Physicians	ePocrates Physicians	ePocrates Penetration
Oncology	5,965	1,005	16.85%
Pulmonology	9,313	1,386	14.88%
Cardiology	23,137	2,991	12.93%
Infectious Disease	5,308	678	12.77%
Emergency Medicine	26,425	3,291	12.45%
Nephrology	5,980	716	11.97%
Endocrinology	4,812	554	11.51%
Gastroenterology	11,360	1,235	10.87%
Neurology	13,077	1,389	10.62%
Neurosurgery	5,962	580	9.73%
Anesthesiology	42,791	4,028	9.41%
Otorhinolaryngology	11,764	1,057	8.99%
Internal Medicine, General	125,390	11,000	8.77%
Allergy	4,666	390	8.36%

Average User: 41 Years Old

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Rules for Developing Apps

Don't Forget the Doctor!

The Acid Test:

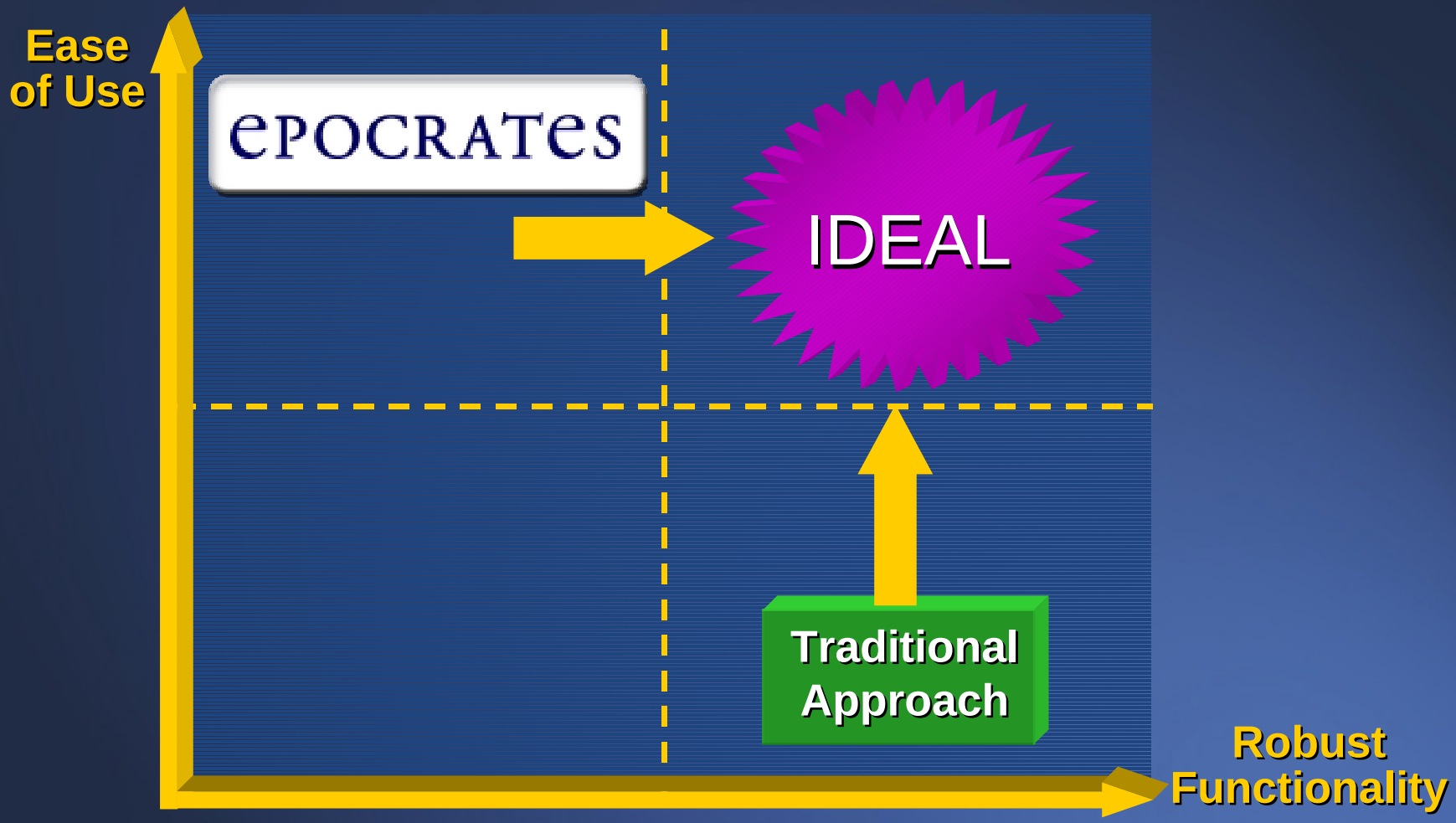
- #1: Does it save me time?
- #2: Does it make me money?
- #3: Does it help me do my job better?
- Need at least 2 of the 3, especially #1

Layman's example:

- Expense Report Tracking

Clinical apps will be first

Development Tradeoffs for MDs



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Climbing the Technology Ladder

Maximize Adoption Then Build Functionality



CRO Study Criteria
Online CME

2001

Renewal Rx
Market Research
ID Reference

2000

One-to-one Messaging
Formulary Reference
Drug Reference
Medical Content Channels
Address/Contact Lists

Today

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ePocrates Partnerships

Particularly Critical in Healthcare

Sponsors

Lilly

PCS Health

ConnectiCare

Connectivity

CIGNA

AdvancePCS

CAREMARK

Content Providers

The New England Journal of Medicine

MDConsult

drugstore.com

UpToDate

Journal Watch

American Academy of Pediatrics

Technology

palm

certicom

AvantGo

Connecting the Hands of Health

Scott Stuewe
Senior Enterprise Manager
Cerner Corporation

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Agenda

- Instructive point-of-care product analogy
- Palm OS® and pull
- Health System and push
- Three Kinds of Applications
- Cerner Corporation’s Core Business
- Cerner Corporation’s Mobility Strategy
- Utopia—where is that anyway?

Analogy: Selling Products to Teens…

Targets:

- Parents—they have much of the money
- Schools—a congregation place
- The Kids—they KNOW what they want
- What works best: Pull with push subsidy…

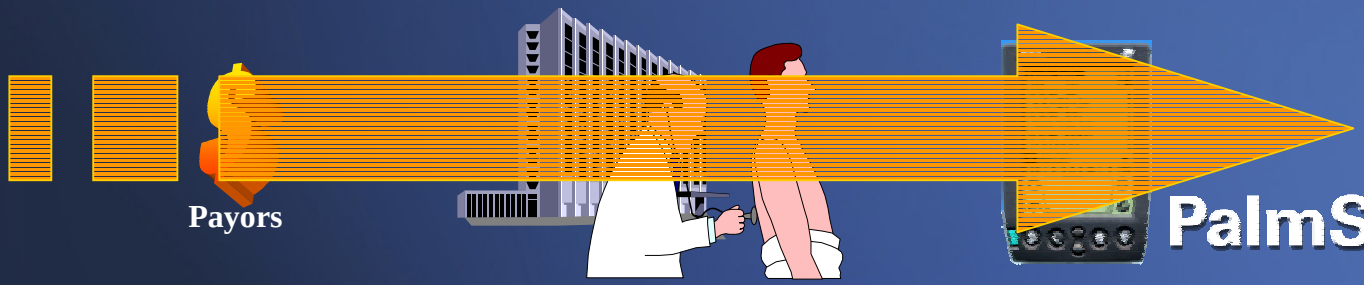


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Doctors and Other Providers

The Analogy:

- Insurance companies are parents—with the money
- The Health System—the community hospital—is the school
- The providers know what they want
  - Don't change the way I practice
  - Don't charge me a lot of money (subsidize me)
  - Acknowledge that I work in a variety of venues



Mobility Push—Health System

Get in Front of Transactions to See Benefits:

- Some would-be point-of-care applications are mobile applications—period
- ADEs cost 100k lives and 76 billion dollars—32% preventable in order entry
- Eligibility and medical necessity not available at point of service
  - Slows AR and drive up costs
  - 30% of testing un-reimbursed
- ***IF the provider's won't use it, all benefits bets are off***

Pull From the Provider

"Don't Make Me Change the Way I Practice"

- Palm has overwhelming market share with the provider
  - Simple and intuitive metaphors
  - Price-point sufficiently low
  - Personal applications co-exist with mandated and business applications
  - Good Conduits available for synchronization
  - Long Battery life
  - Truly pocket-sized
  - Most applications available
  - Low cost, reliable wireless network

Healthcare Mobility Applications

- Content Applications
  - Content Managed by a non-mobile application
  - Example—Pocket PowerChart
  - Focus is on making content mobile
- Companion Applications
  - Complete the work-flow of traditional apps
  - Depend on parent application
  - Examples
- Connected Applications
  - Two-way in both Synch and Wireless
  - Mobile app has stand-alone utility

About Cerner Corporation



- 20 years in HIT
- Millennium architecture proven
  - Large scale health systems—UIC, DMC, Mayo
  - 720 Applications in production
  - 116 Clients of the new architecture
  - Representing 189 sites
- Just announced HNAObjects Partnership
- Virtmed is our HNAObjects Alpha

HNA Millennium Product Breakdown

1 Architecture

HNA Millennium

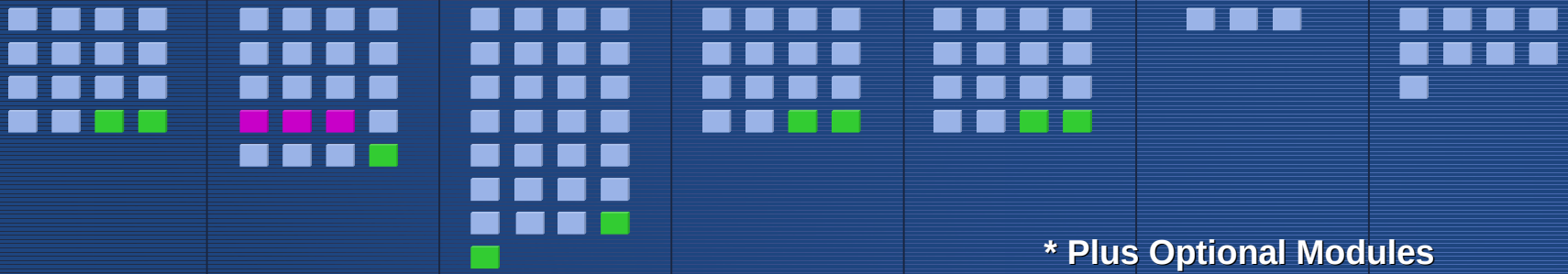
7 Solution Groups

Enterprise Systems & Repositories	Clinical Systems for Direct Care	Clinical Systems for Care Centers	Financial and Operational Mgmt Systems	Decision Support & Knowledge Solutions	Interface Technologies	eHealth (IQHealth)

29 Product Lines (32)

Access Mgmt (4)	Nursing (8)	Laboratory (9)	Patient Acct (3)	Discern (3)	Open Engine (1)	Personal Health(3)
EMR (4)	ICU (1)	Radiology (4)	Agrmnt Mgmt (1)	Structured Doc (2)	Open Port (2)	Phys Conn (3)
OMF (4)	Physician (3)	Pharmacy (3)	Medical Mgmt (4)	Content (3)		Demand Mgmt (2)
EEM (4)	Home Health (7)	Emergency (2)	HIM/CDI (3)	Health Facts (6)		Population Hlth(2)
		Surgery (4)	Materials (3)			
		Cardiology (4)	Equipment (1)			
		Obstetrics (1)				
		Oncology (1)				

100 Applications (109)



* Plus Optional Modules

Cerner Corporation's Mobility Strategy

*Applications for **Health System Buyer—for All Users***

- Content—Pocket PowerChart—Now Available
- Companion Applications—Lab Orders and results—ePathLink Companion—Q2 2001—Others Q3
- Connected Applications—new version of Pocket PowerChart in cooperation with Virtmed—2001



Cerner Corporation's Mobility Strategy

Virtmed—Hand-held Applications and Infrastructure Services Partner

- Focus on the health-system
- Joint Product management—deep joint learning
- Applications are integral to our clinical applications
- Virtmed Services and applications for our clients



Cameron Lewis
SVP Marketing & BD
cameron@virtmed.com

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VIRTMED Background

- Founded in 1998
  - MIT engineers and Harvard physician
- Developed and deployed charge capture
  - 3 Institutions using
- Acquired PatientKeeper
  - 43,000 downloads, 22,000 paid
  - Mostly in-patient users
- 15 installations by EoQ1/01
- CERNER partnership September/00
- Lab Transaction ALPHA October/00

Key Learning From Installations

- Institutions and users want one PDA for all applications
- No application has emerged as The KILLER APP
- Must forge strong ties with HIT group
- Adoption of personal technology by doctors declines starting immediately after introduction and approaches ZERO after 6 months
- NO incentives or measurable value
  - No sustained adoption

Creating a Solution

- KILLER APP may be a platform
- CERNER relationship helps provide platform environment
- Supply chain and workflow issues remain open
- Value chain and payment for services unclear
- Patient and provider identification, authorization, verification, non-repudiation difficult
- Selling a tool for saving time, money and lives


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Resources

PalmSource Sessions

- Healthcare Deployment 809P
  - Today, 1:30–2:30, Room 203/4
- Healthcare Connections 810R
  - Today, 2:45–3:45, Room 209/10

Websites

- www.palmos.com
- www.mohca.org
- www.pdamd.com
- www.handheldmed.com

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